



INTRODUCTION

Understanding the Scope of Drone Forensics

Investigation Context

Drone forensics examinations typically arise in critical cases involving **border surveillance** , **terrorism** , **privacy violations** through illegal surveillance, and **accidents** involving civilian or commercial drones.

Core Competencies Required

- Digital evidence handling and preservation
- Embedded systems analysis
- Wireless communication forensics
- Legal and procedural compliance

OBJECTIVES

Primary Objectives of Drone Forensics Investigation



1. Identify Drone Model

Analyse serial numbers on frame, motors, battery, and controller. Examine firmware signatures and hardware identifiers to trace manufacturer origin.



2. Trace the Operator

Correlate mobile application logs, registered user accounts, controller IDs, paired devices, and Wi-Fi/RF communication artifacts to identify the pilot.



3. Reconstruct Flight Path

Utilise GPS telemetry logs, altitude and speed data, takeoff and landing coordinates to recreate complete flight trajectory and operational patterns.



Data Recovery and Intent Determination

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4. Recover Stored Data

Extract images, videos, flight logs, system event logs, and cached telemetry data from onboard storage devices including SD cards and internal memory.

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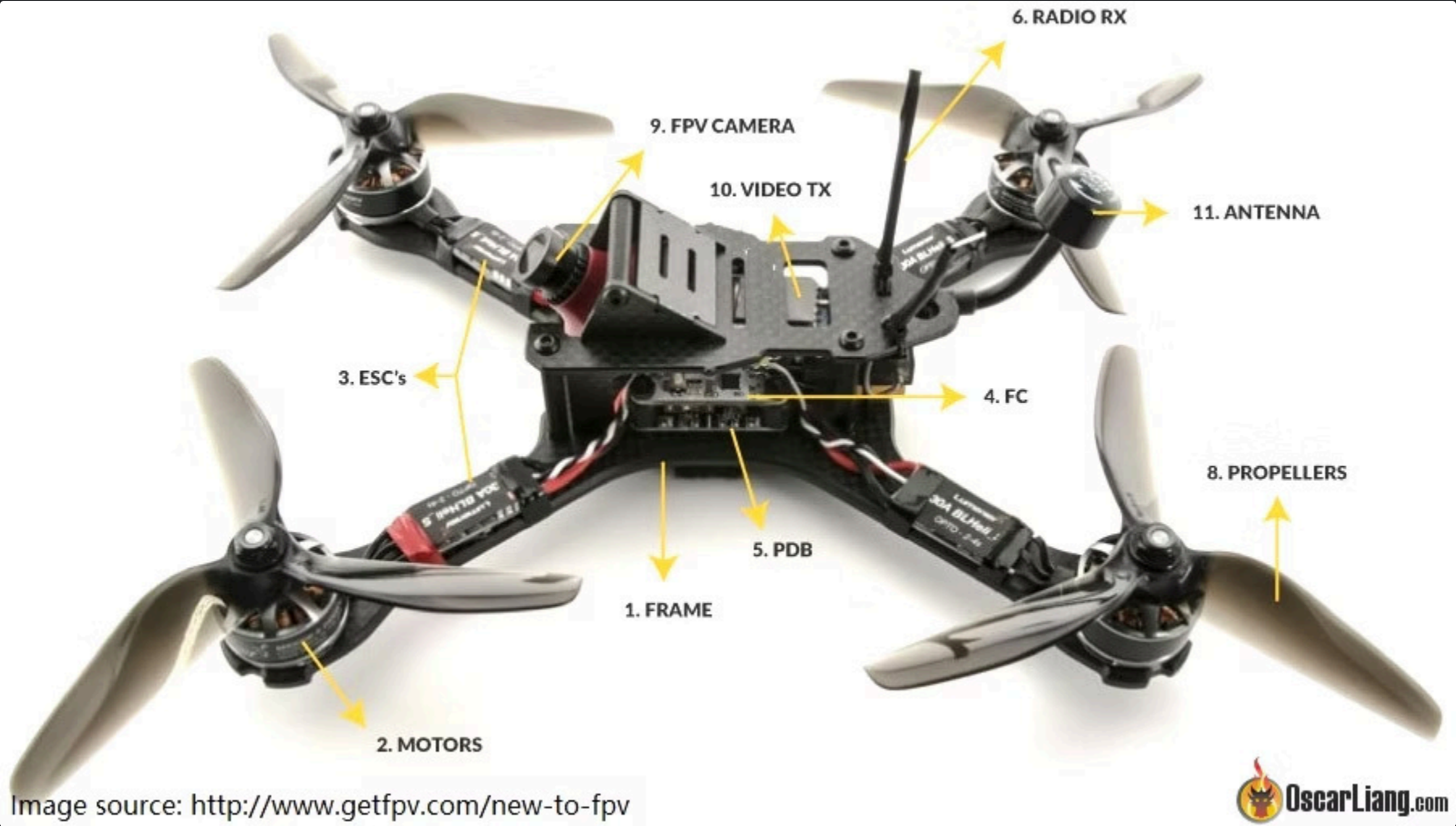
5. Determine Intent and Usage

Correlate all recovered evidence to establish whether the drone was used for legitimate recreational or commercial purposes, or for espionage, surveillance, or illegal payload delivery.

Real-World Case Study: In multiple border-area incidents in India, recovered drones were identified as commercial quadcopters by matching motor and battery serial numbers, enabling authorities to trace the supply chain and country of origin successfully.

Critical Hardware Components – Forensic Perspective

Understanding internal components is essential for evidence correlation and root-cause analysis in drone forensic investigations.



Frame/Body	Damage analysis helps reconstruct crash dynamics and impact patterns
Motors & Propellers	Wear patterns indicate flight duration and operational history
Battery (Li-Po)	Charge cycles reveal operational history and usage patterns
Flight Controller	Stores critical configuration data and comprehensive system logs
GPS Module	Essential for location tracking and complete route reconstruction
Camera & Gimbal	Primary source of surveillance evidence and visual data capture



Software Components and Advanced Analysis

Firmware Analysis

Contains manufacturer signatures, flight restrictions, geofencing data, and operational logs. Firmware examination may reveal geofencing bypass attempts, tampering evidence, or custom modifications indicating malicious intent.

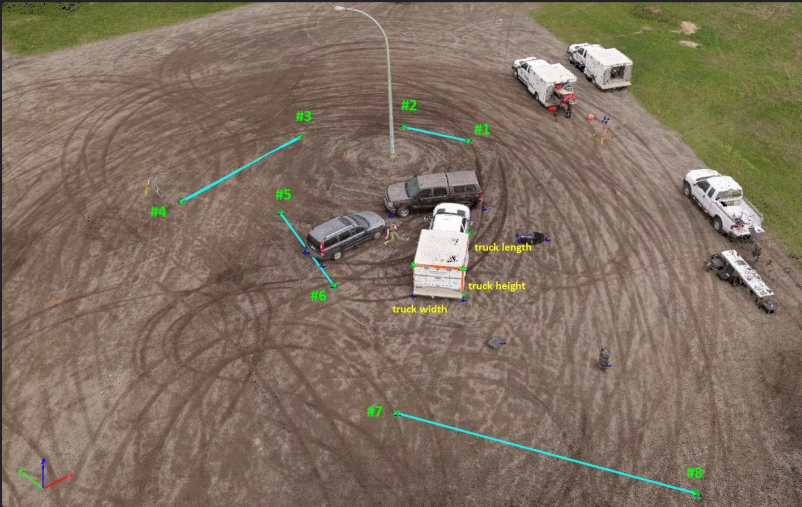
Mobile Applications

Store critical pilot identity information, complete flight history, cloud synchronisation data, and registered user account details essential for operator identification.

Ground Control Software

Used in professional and military operations for mission planning, waypoint configuration, and advanced operational parameters in sophisticated drone deployments.

Physical Evidence Collection and Analysis



Types of Physical Evidence

- Drone airframe (damaged or intact)
- Motors, propellers, and batteries
- SD cards and onboard storage media
- Camera and gimbal units
- Remote controller devices
- SIM cards (LTE-enabled drones)
- Payloads (contraband, surveillance devices)

❏ **Critical Handling Protocol:** Always photograph and document the scene before collection. Use anti-static and Faraday bags. Never power on devices to prevent log overwriting. Strictly follow chain of custody procedures for legal admissibility under cyber laws.

Strategic Importance in Modern Investigations



National Security

Critical for counter-terrorism operations, border surveillance analysis, and preventing security breaches through unauthorised UAV operations in restricted zones.



Cyber-Physical Crime Investigation

Addresses complex investigations involving both digital artifacts and physical evidence in smuggling, espionage, and illegal surveillance operations.



Legal Compliance

Ensures digital evidence meets admissibility standards under cyber laws, maintaining proper chain of custody and forensic integrity throughout investigations.



Intelligence Gathering

Advanced incident response capabilities and intelligence gathering techniques essential for proactive threat detection and criminal network mapping.

